GRADE 3 · INTRO PYTHON TURTLE

Read the Turtle Code

Ontario Math C3.2 — read code and predict

Name: .... Date: ....

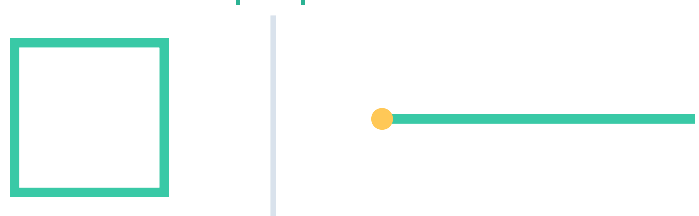
★ I can read a turtle loop and predict what Bit draws.

Picture it first

Good coders read code and picture what it does BEFORE running it. For a turtle loop, look for two clues. Clue 1: is there a turn? A turn makes a shape; no turn makes a straight line. Clue 2: how many sides? The range number tells you.

Read each loop below and predict Bit's drawing. Then you can run it to check.

read the loop → picture what Bit draws



with a turn
→ a shape

no turn
→ a straight line

A turn makes a shape; no turn makes a straight line.

Loop A — read it

PYTHON

```
for i in range(4):
    bit.forward(30)
    bit.right(90)
```

1 Read Loop A

Loop A has a turn (right 90) and range(4). What shape does Bit draw, and how many sides? ______

____

2 Read another loop

Loop B is: for i in range(3): bit.forward(40); bit.left(120). What shape does Bit draw, and how many sides?

3 Predict, then run

Loop C is: for i in range(5): bit.forward(20) (no turn). PREDICT what Bit draws and how long it is in total. Then run it.

4 Challenge

Loop A draws a square with range(4), right 90. After the loop finishes, WHERE is Bit — back at the start, or somewhere else? How do you know?

Big idea

You can read a loop and picture the drawing: a turn means a shape, no turn means a line, and the range number tells you how many sides.

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Read the Turtle Code

Quick facilitation notes & answer key

What this worksheet teaches

Good coders picture what code does before running it. Students read several repeats and predict each drawing: a turn makes a shape, no turn makes a straight line, and the repeat number gives the number of sides. They also reason about where Bit ends up. No coding experience needed.

Before you start — showing the drawing (optional)

The worksheet shows the typed instructions AND a picture of each drawing, so you can teach straight from the printout — no computer required.

To show it live (optional): open a browser, go to a free "Python turtle" playground (e.g. search "trinket python turtle"), type the few lines from the worksheet, and press Run — a turtle draws it. You only press Run.

How to lead it (about 30 minutes)

1. Set the habit: "Read it and picture it first, then run to check."
2. Show it (you do). Read one repeat aloud and ask the class to picture it before you run it.
3. Let them work. Students do Exercises 1–4 (Ex 4 reasons about where Bit ends), solo or in pairs.
4. Wrap up: look for two clues — is there a turn (shape vs line), and how many repeats (sides).

Answer key (these are the verified correct answers)

Exercise 1 — a square; 4 sides.

Exercise 2 — a triangle; 3 sides.

Exercise 3 — a straight line; 100 long (5 × 20).

Exercise 4 (challenge) — back at the start (a closed square returns to where it began).

If students get stuck — and ways to adjust

Watch for: ignoring whether there is a turn — no turn means a straight line, not a shape.

Make it easier: have students point to the turn line (if there is one) before predicting.

Make it harder: ask where Bit ends after a triangle (also back at the start).

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

In plain terms: students write and run step-by-step instructions — including a "repeat" instruction — to make something happen.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions, change one, and explain what the change did to the result.

You'll know it worked when

The student predicts the shape or line from each repeat (Ex 1–3) and reasons that a closed shape brings Bit back to the start (Ex 4).